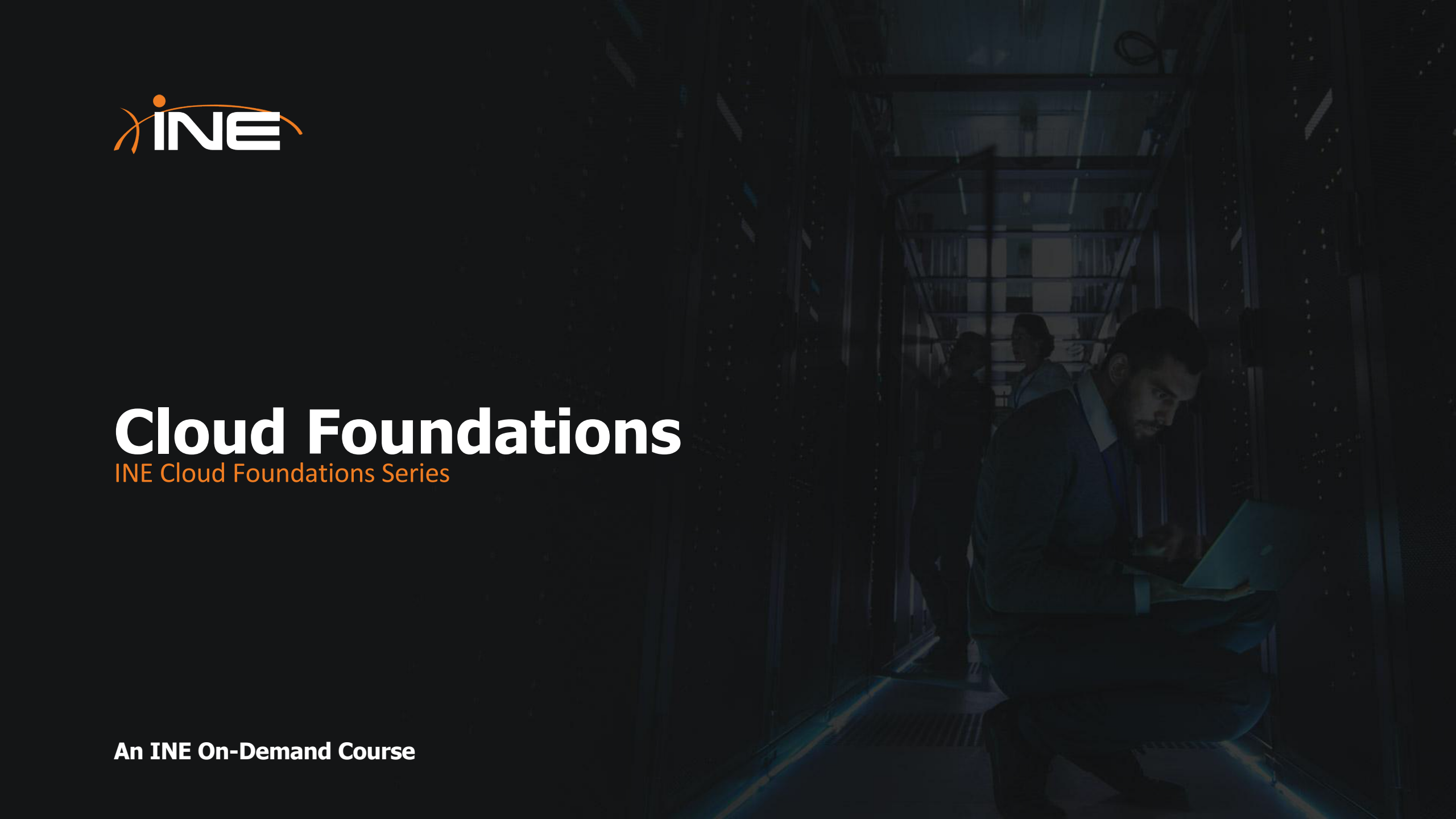


Cloud Foundations

INE Cloud Foundations Series

An INE On-Demand Course

Tracy Wallace

Azure Solutions Architect Expert

twallace@ine.com

@TracyWallaceTec

linkedin.com/in/tracy-wallace-746482a

Topics

Introduction
Cloud Management
Cloud Services
Cloud Providers

Learning Objectives

After completing this course, you will be able to:

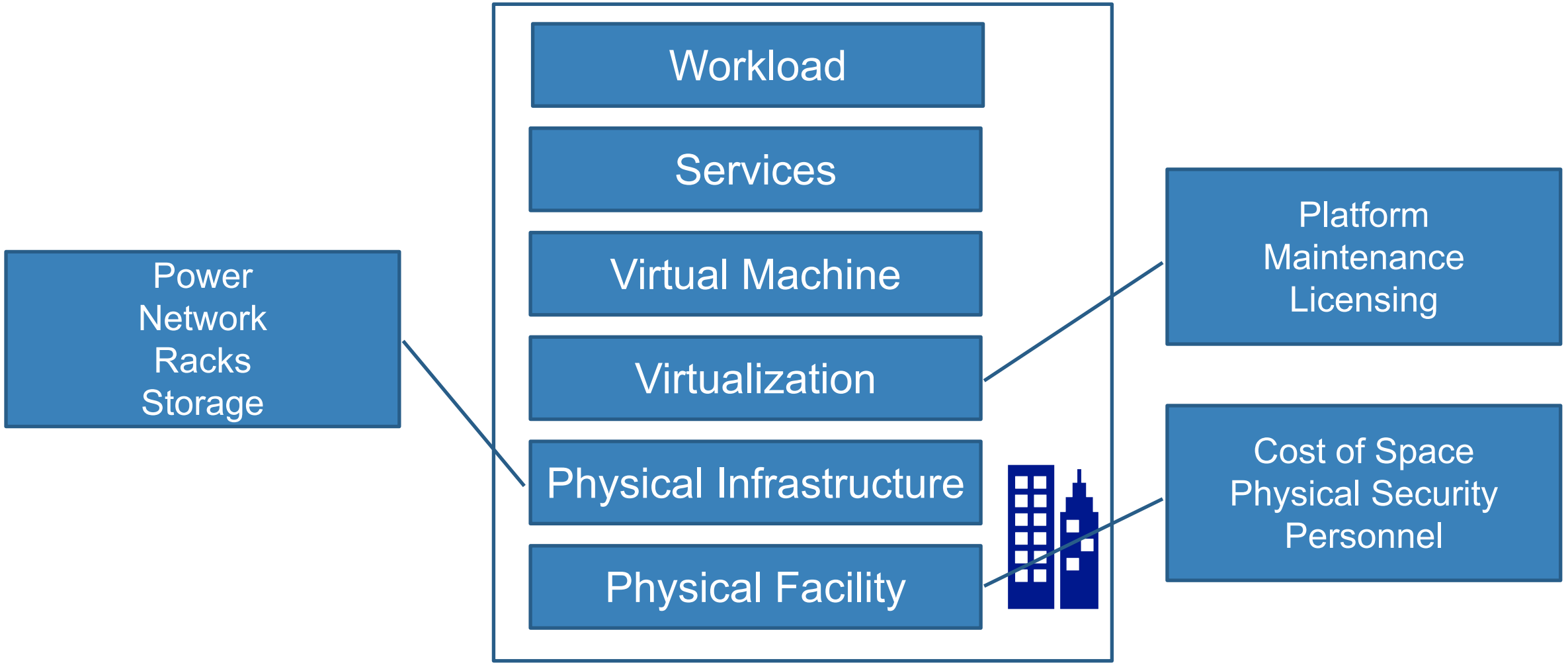
- Define what is meant by "the cloud"
- Explain why organizations are moving workloads to the cloud
- Identify services that are provided in the cloud
- List the major cloud providers

What is the Cloud?

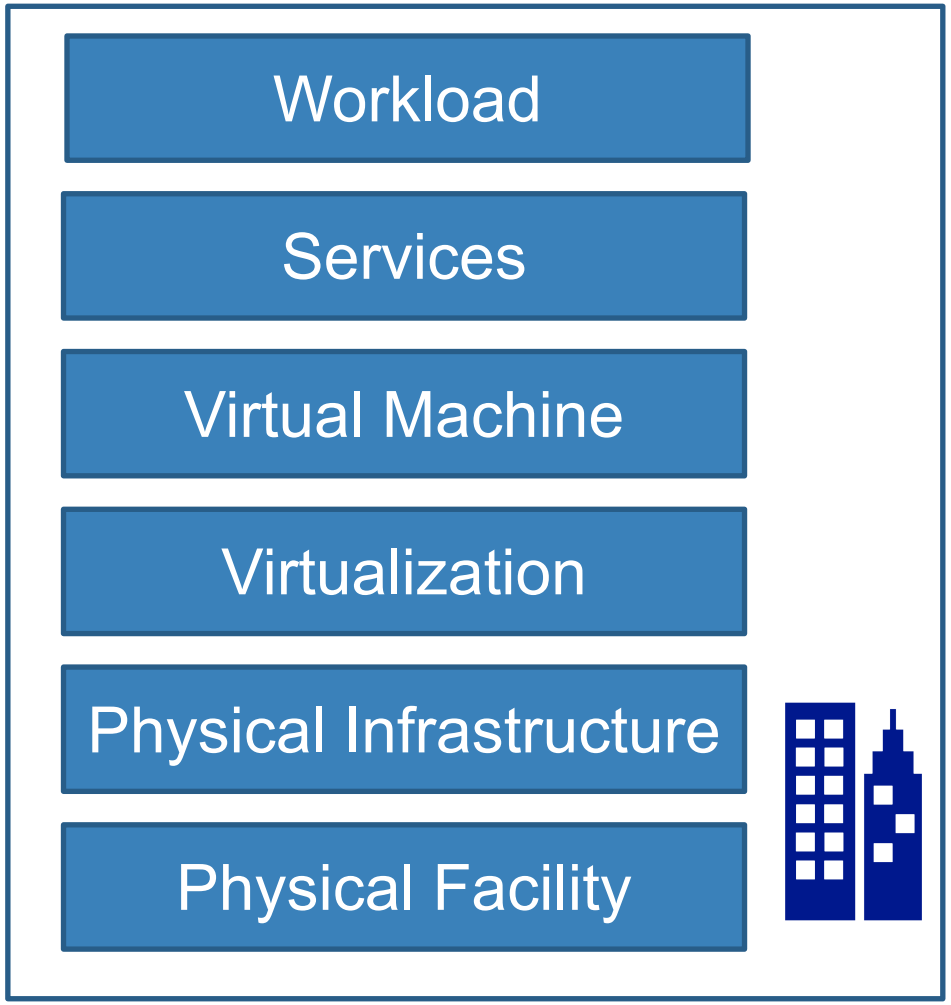
What is the Cloud?

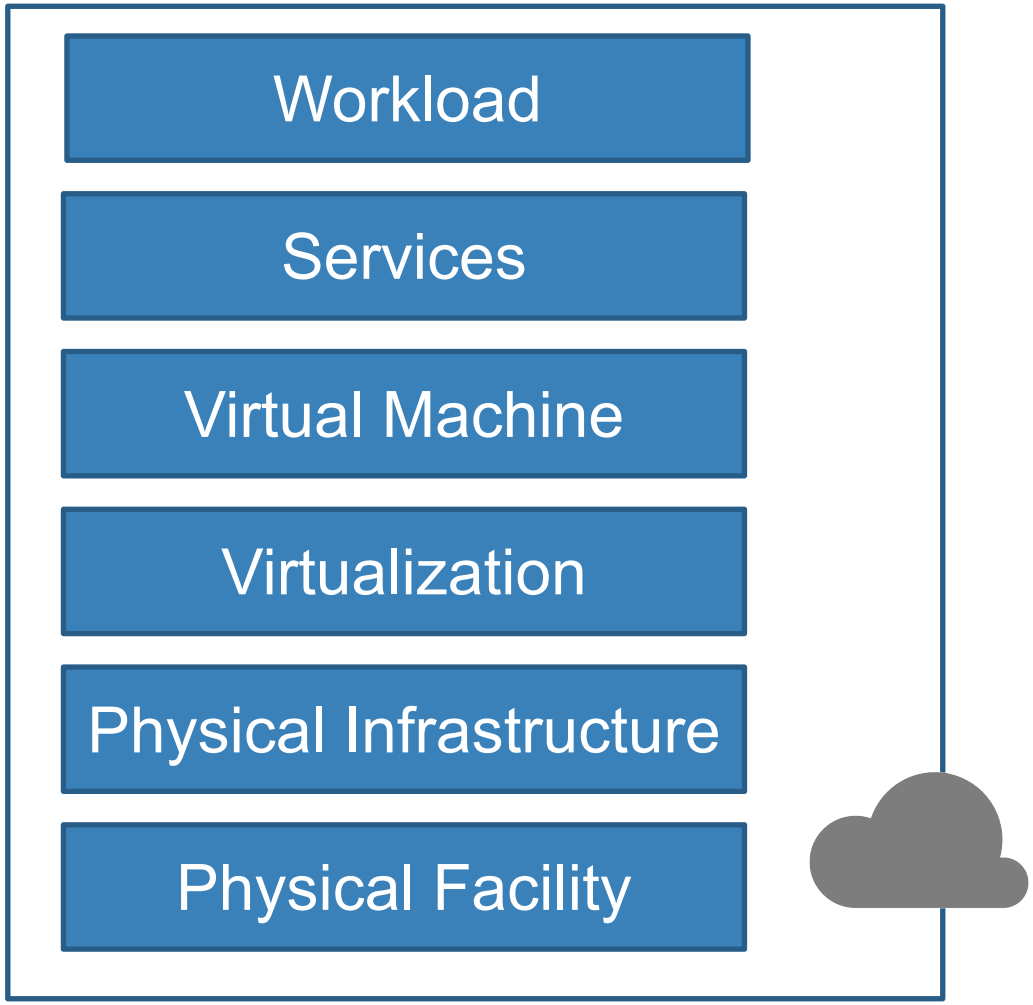
- Cloud Architecture
- Types of Cloud Services
- Accessing Cloud Services
- Demo: Why I Love the Cloud

On-Premises Information Systems Architecture

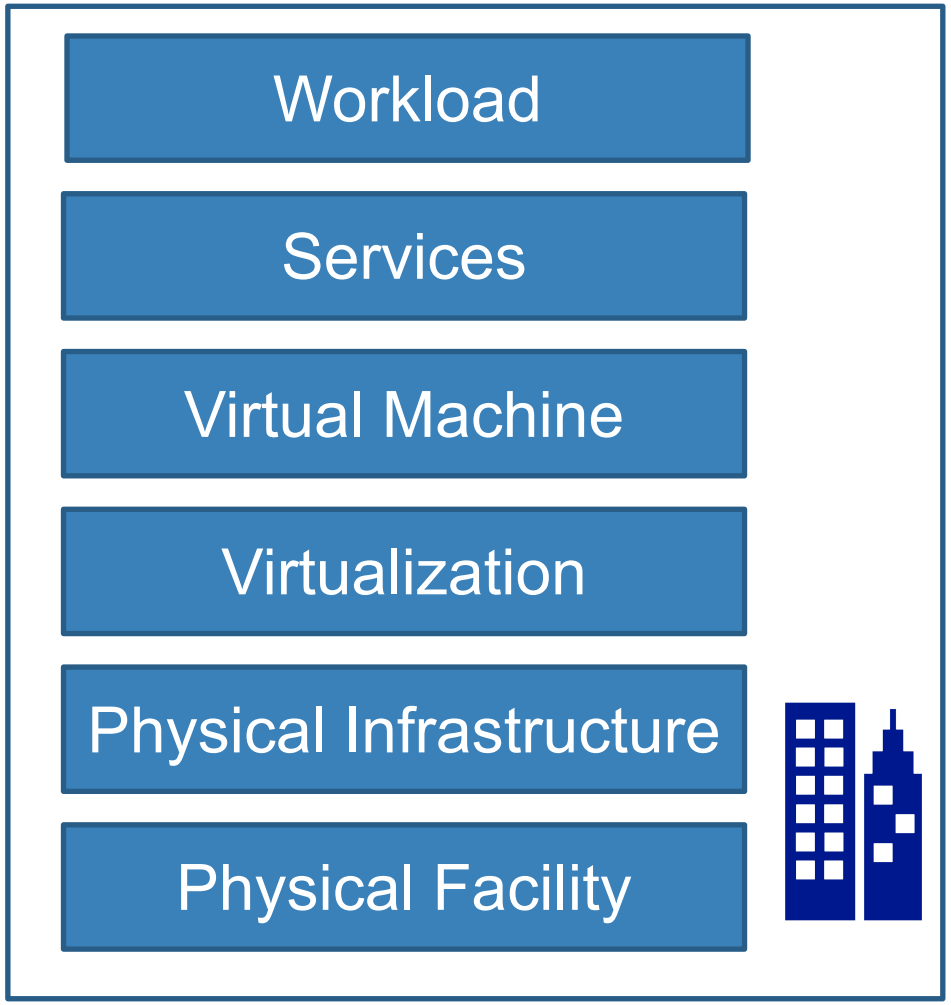


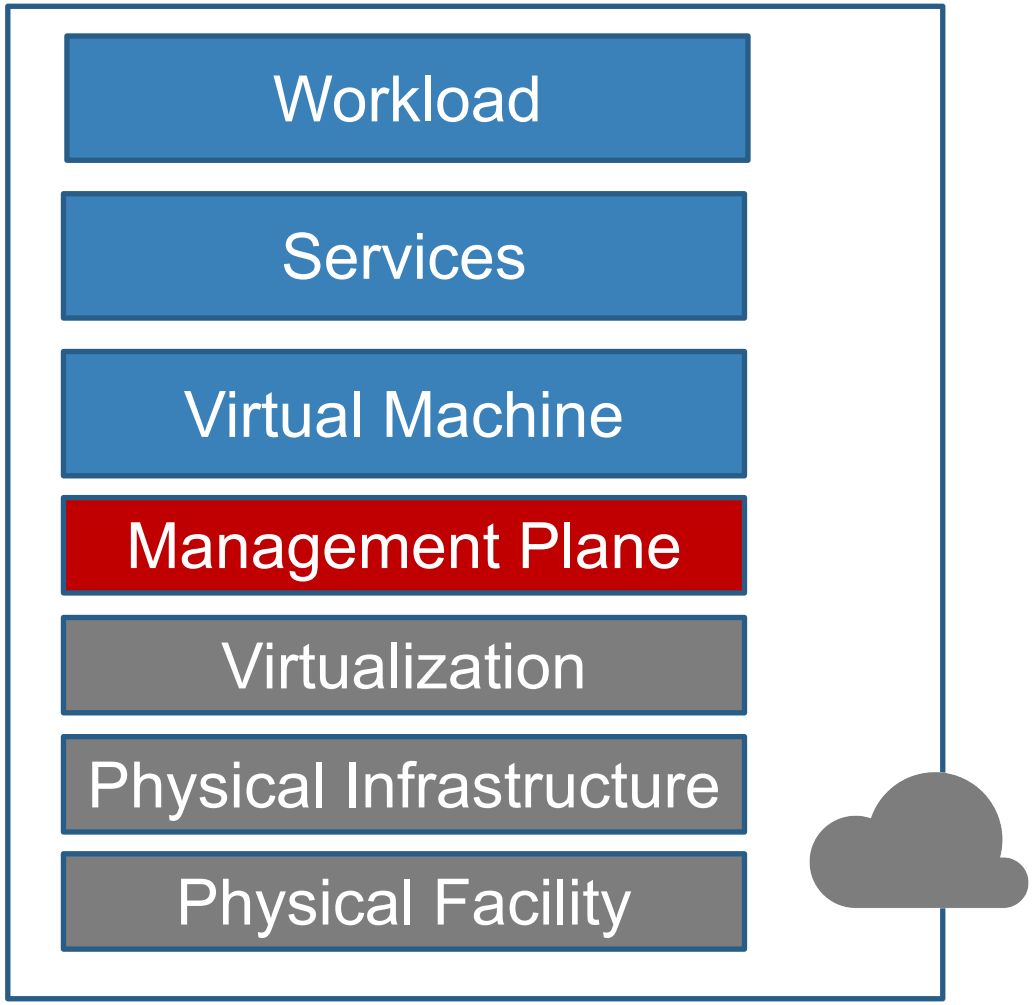
Cloud Architecture



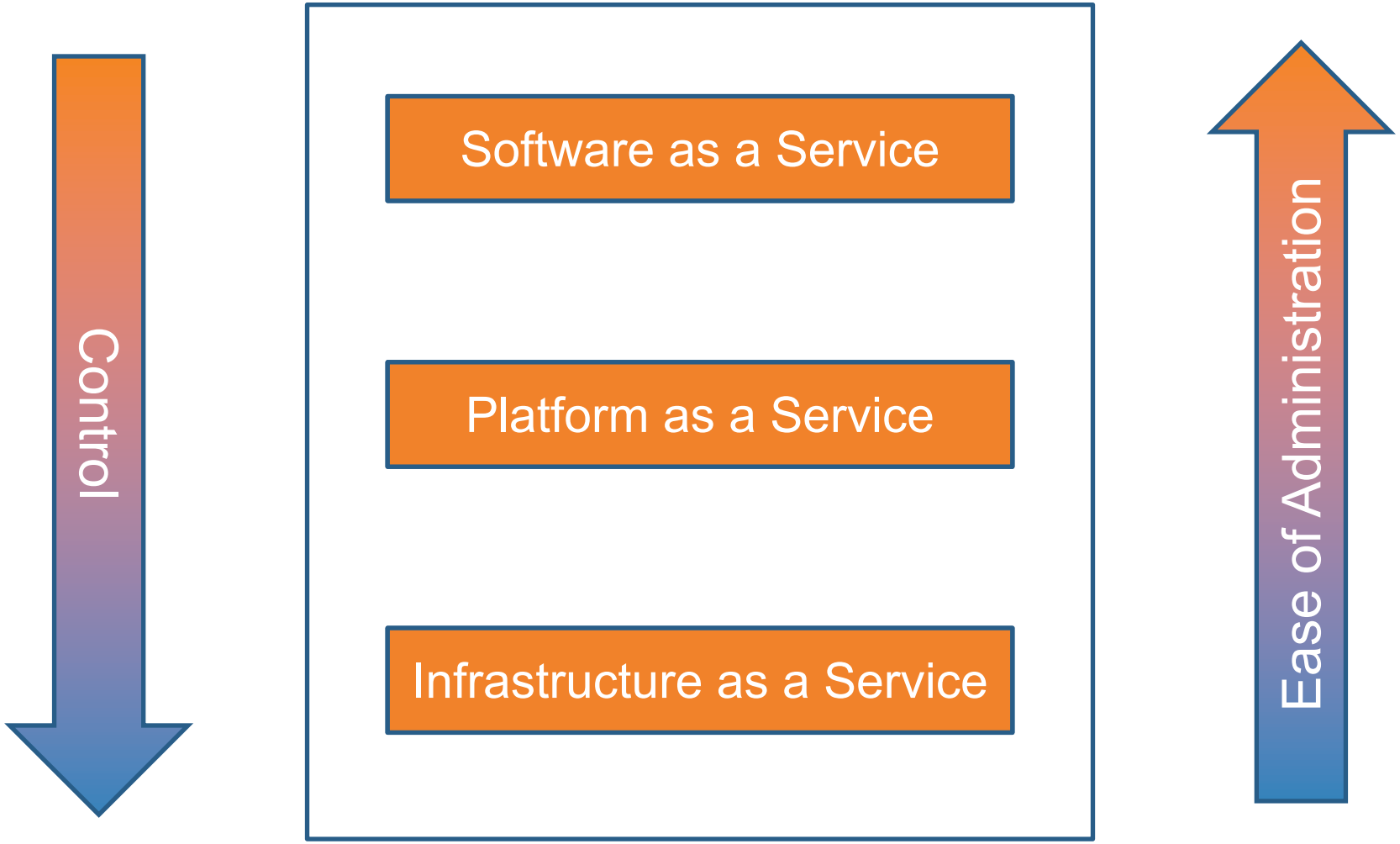


Cloud Architecture

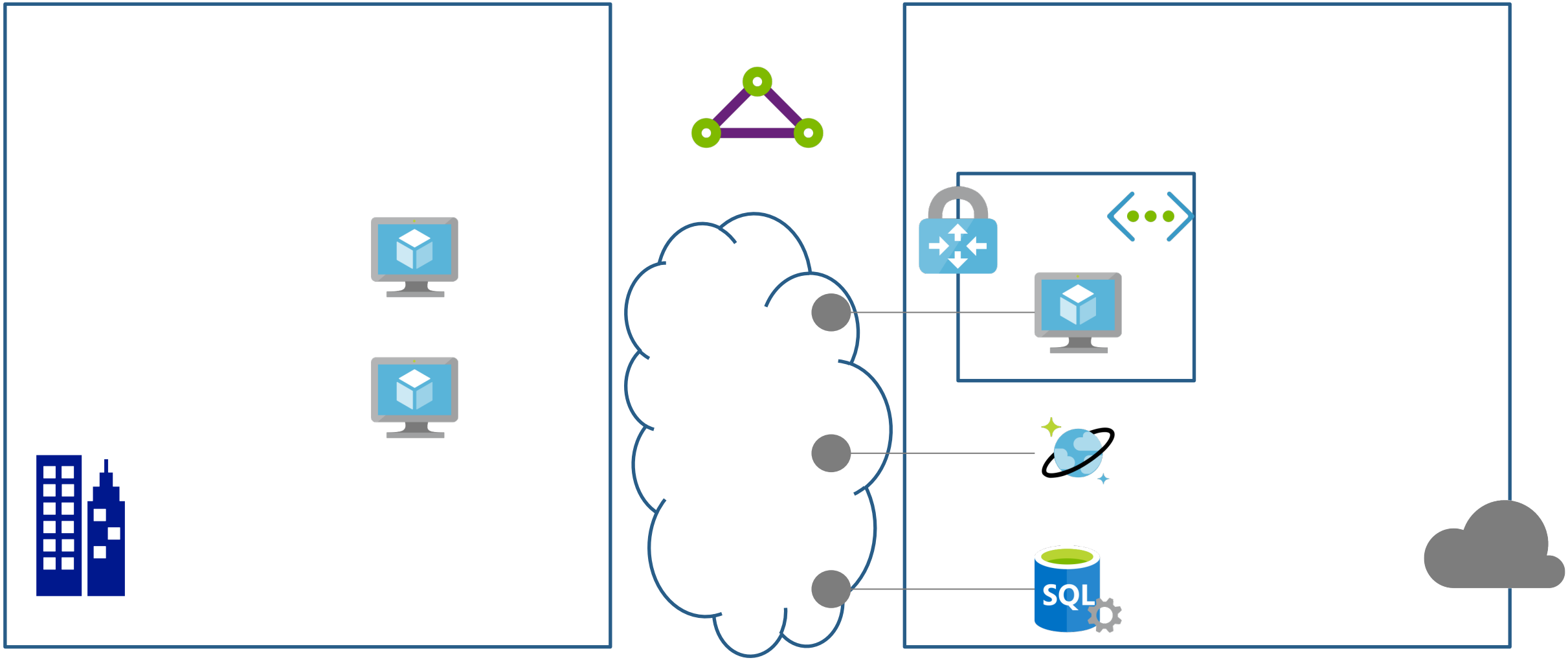




Types of Cloud Services



Accessing Cloud Services



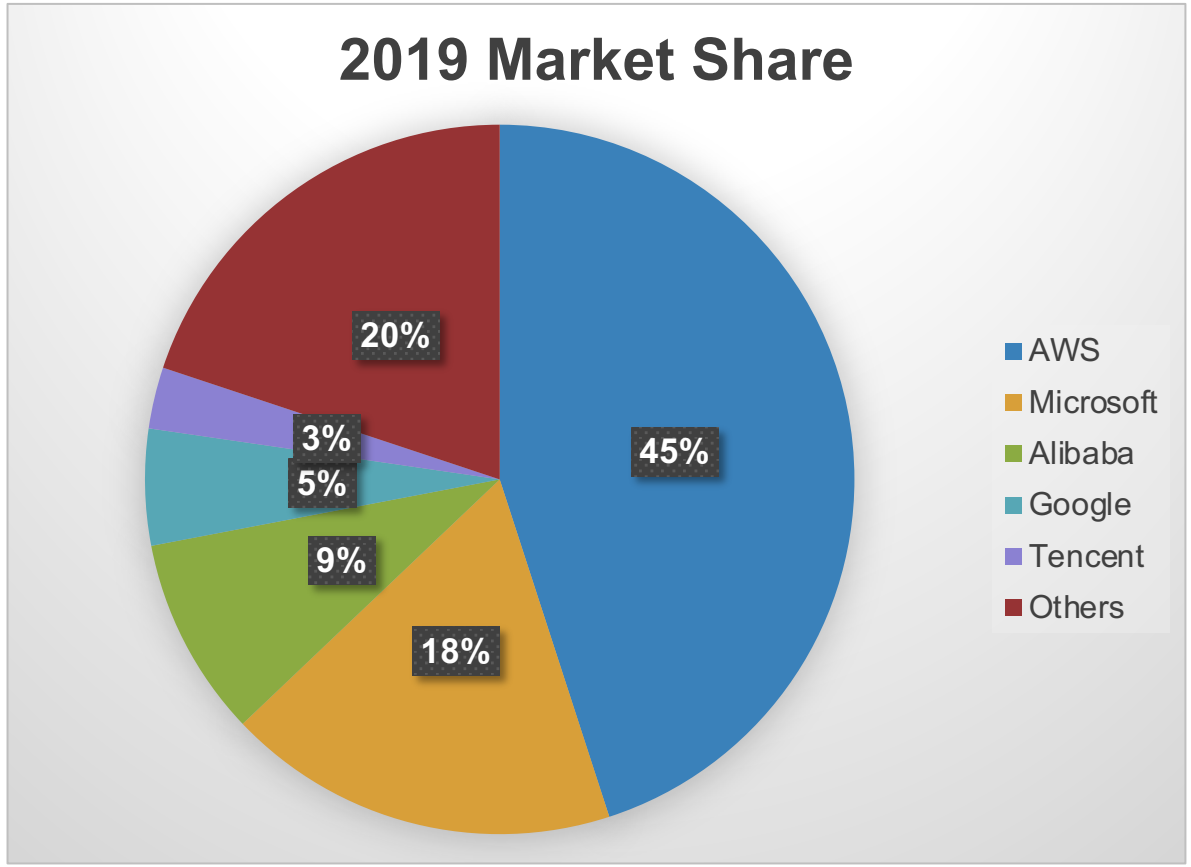
Demo: Why I Love the Cloud

Who are the Cloud Providers?

Who are the Cloud Providers?

- Cloud Market
- AWS
- Azure
- Google Cloud Platform
- Other Cloud Providers

Cloud Market



Year	Estimated market
2019	$216B (US)
2020	$371B (US)
2025	$823B (US)

Source: Various

Source: https://www.gartner.com/

Officially launched 2006

Over 1,000,000 active users

Over 190 countries

18 global regions

Over 175 products and services

Publicly available 2010

Over 169 services

Over 300k customers

57 regions

31 geographies

Google Cloud Storage launched 2010

Over 90 services

Not sure how many users

Google Cloud

Other Cloud Providers

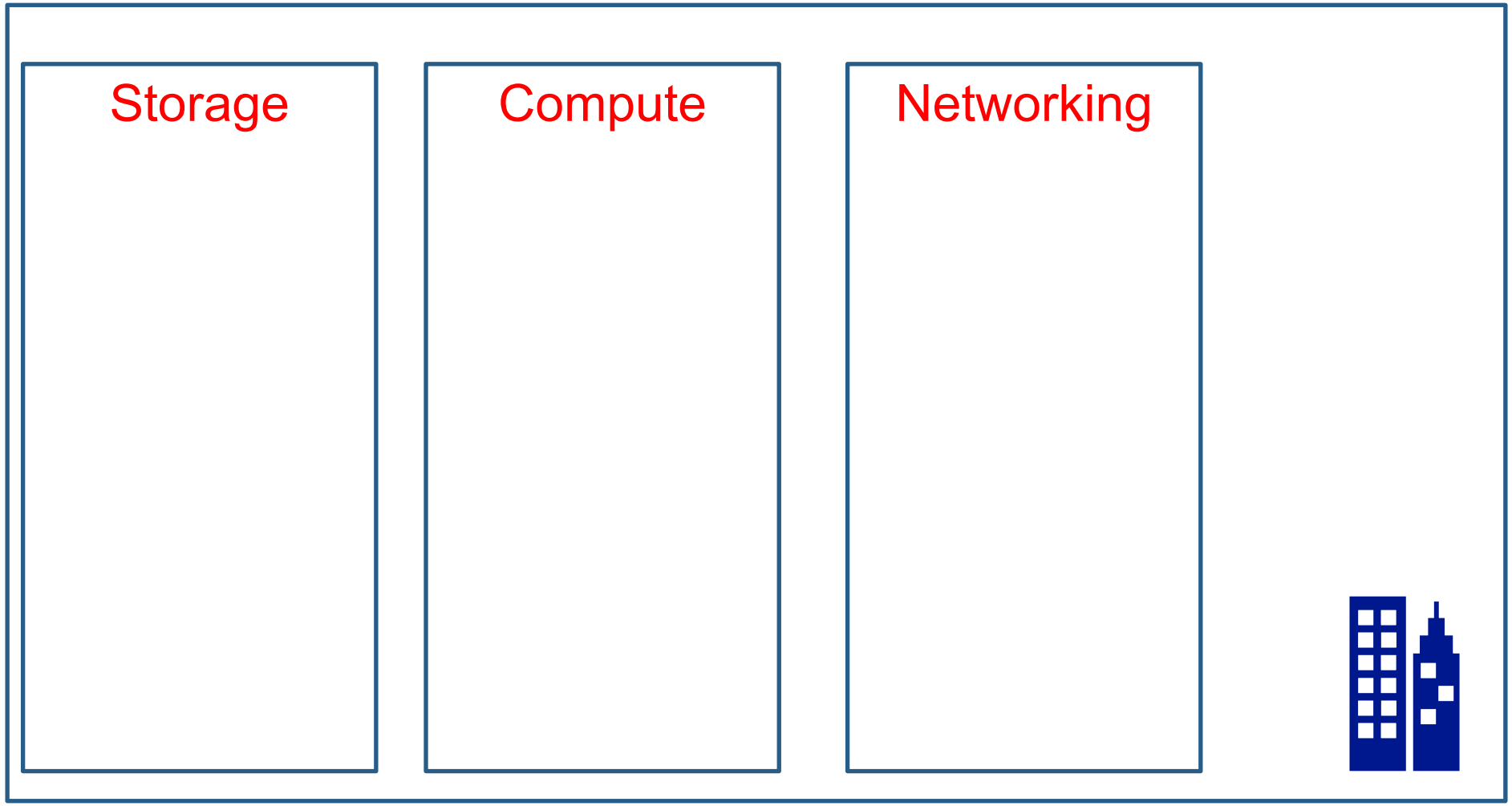
- In the neighborhood of 20% of the market
- Tencent and Alibaba in China
- IBM and Oracle
- Digital Ocean
- Private cloud providers

Why Choose the Cloud?

Why Choose the Cloud?

- Cloud Economics
- CapEx vs. OpEx
- Consumption-Based Spending
- Functional Advantages
- Also, Maybe Not

Cloud Economics



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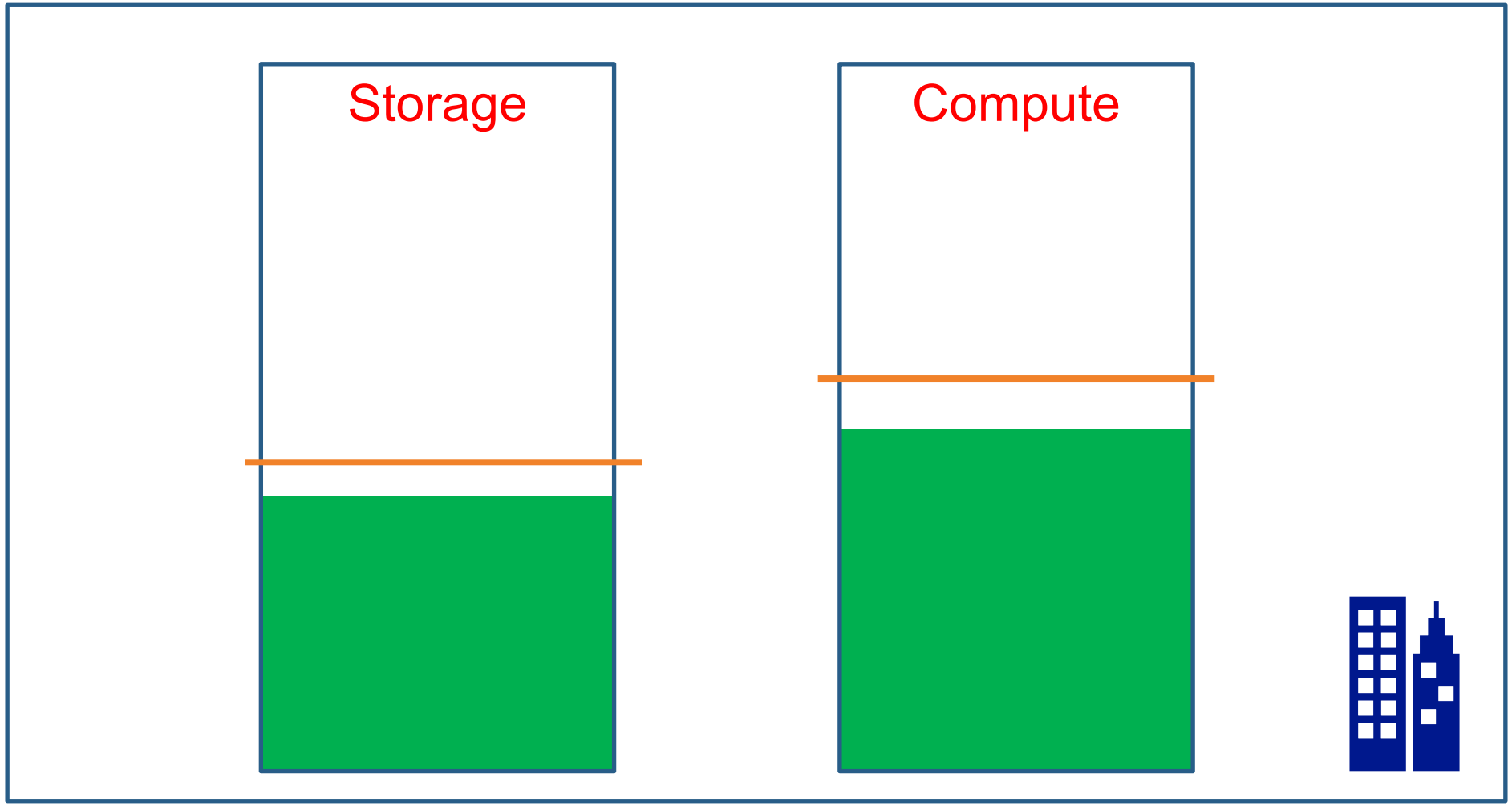

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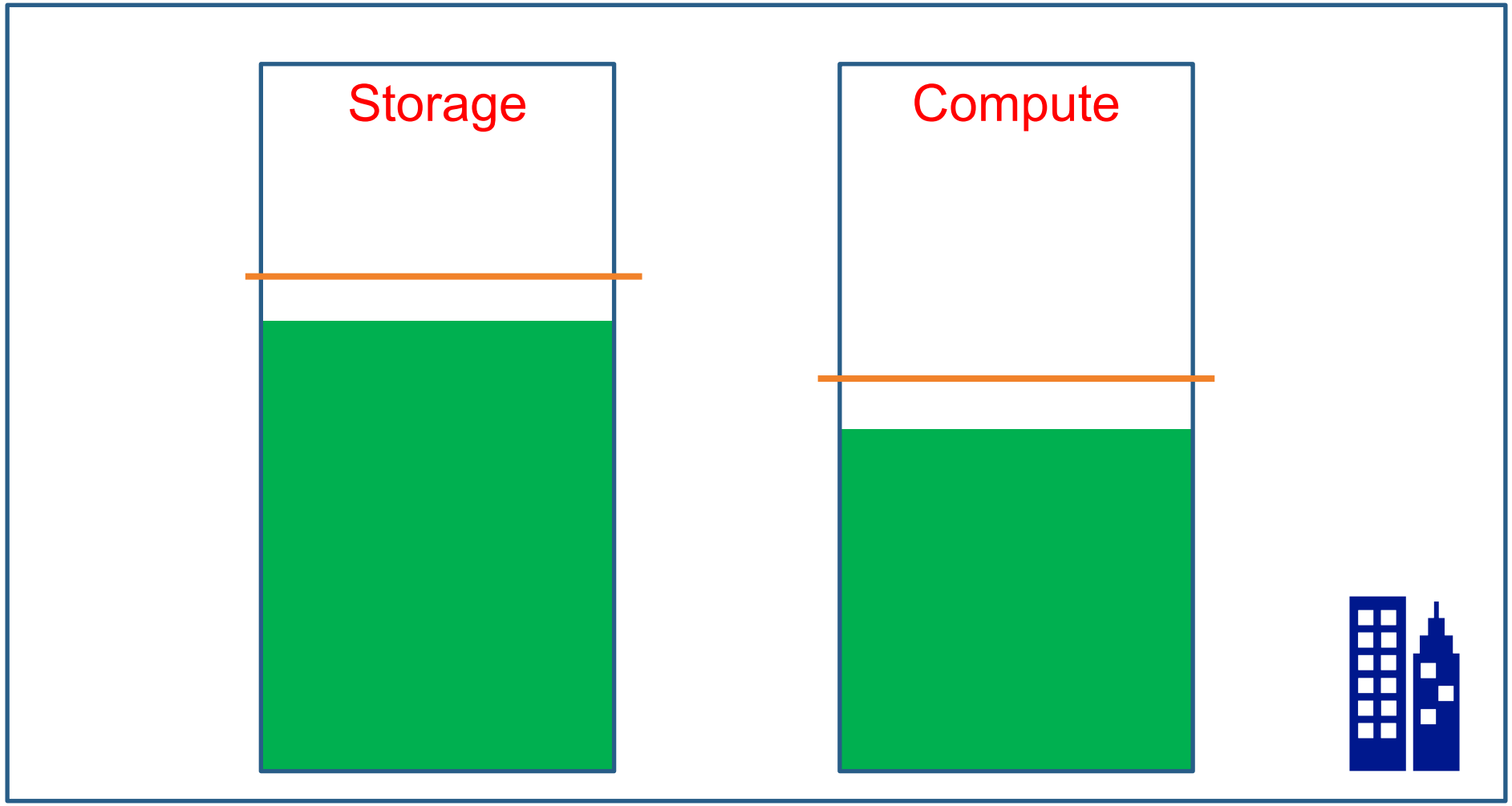

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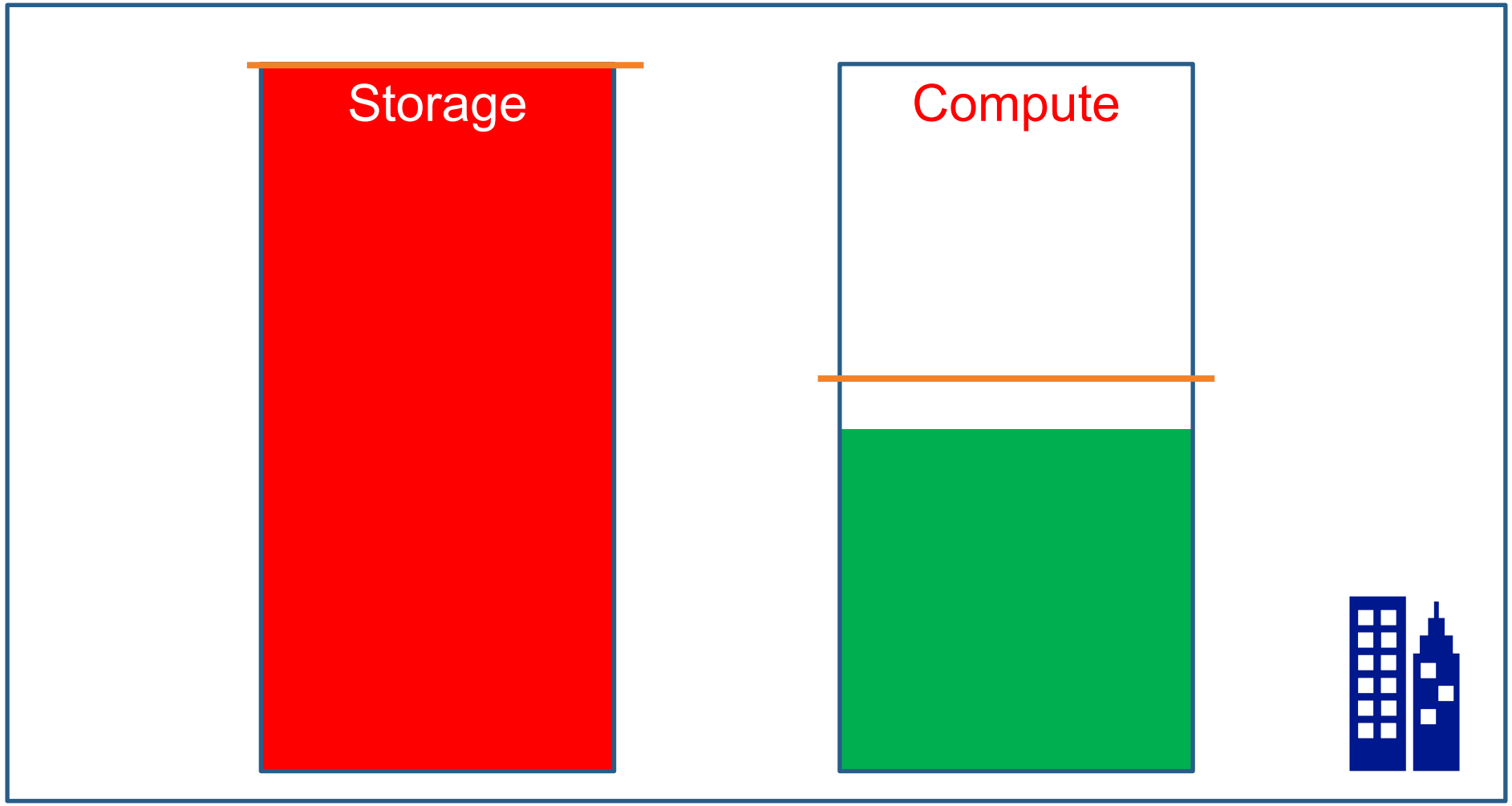

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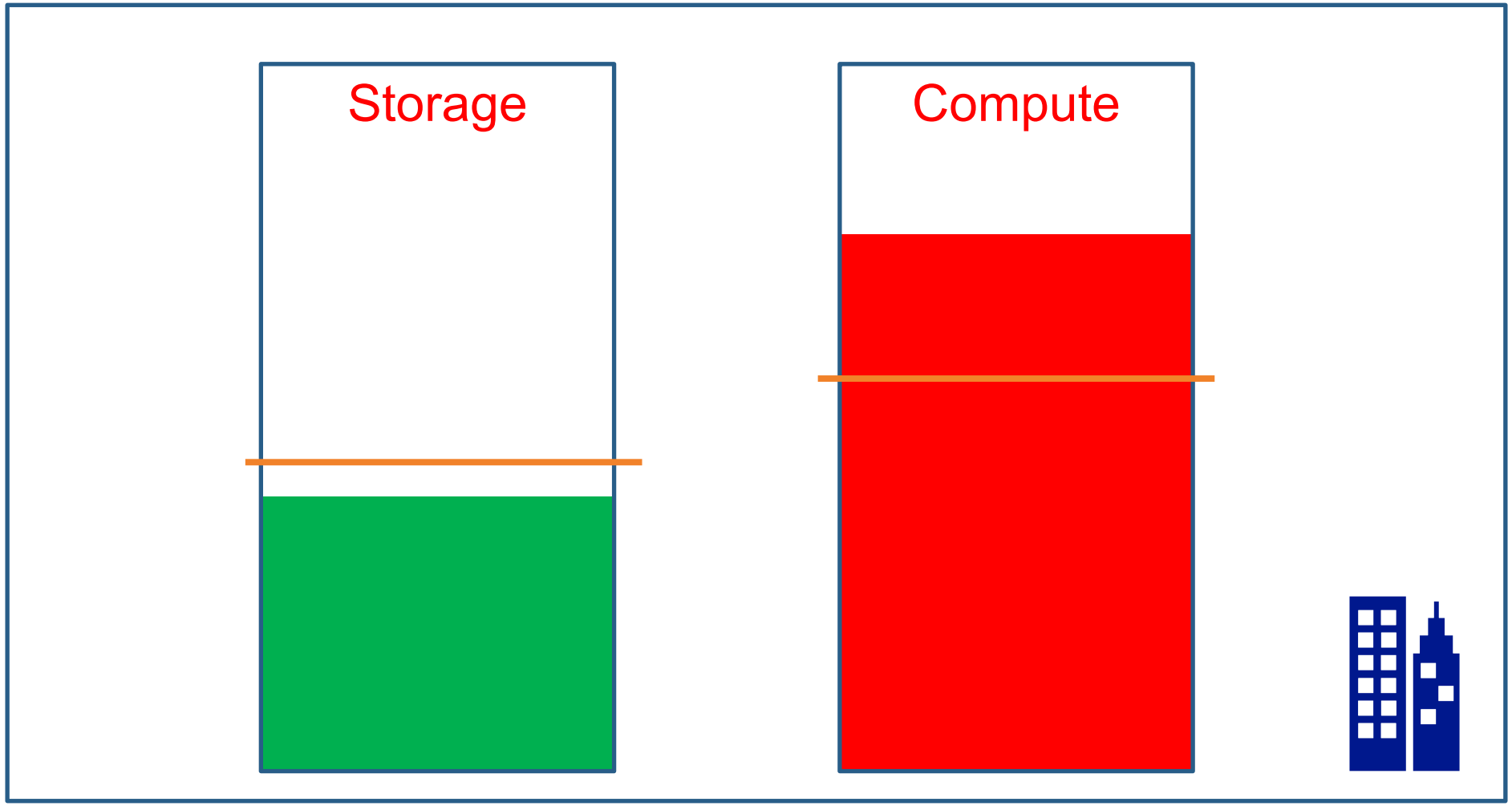

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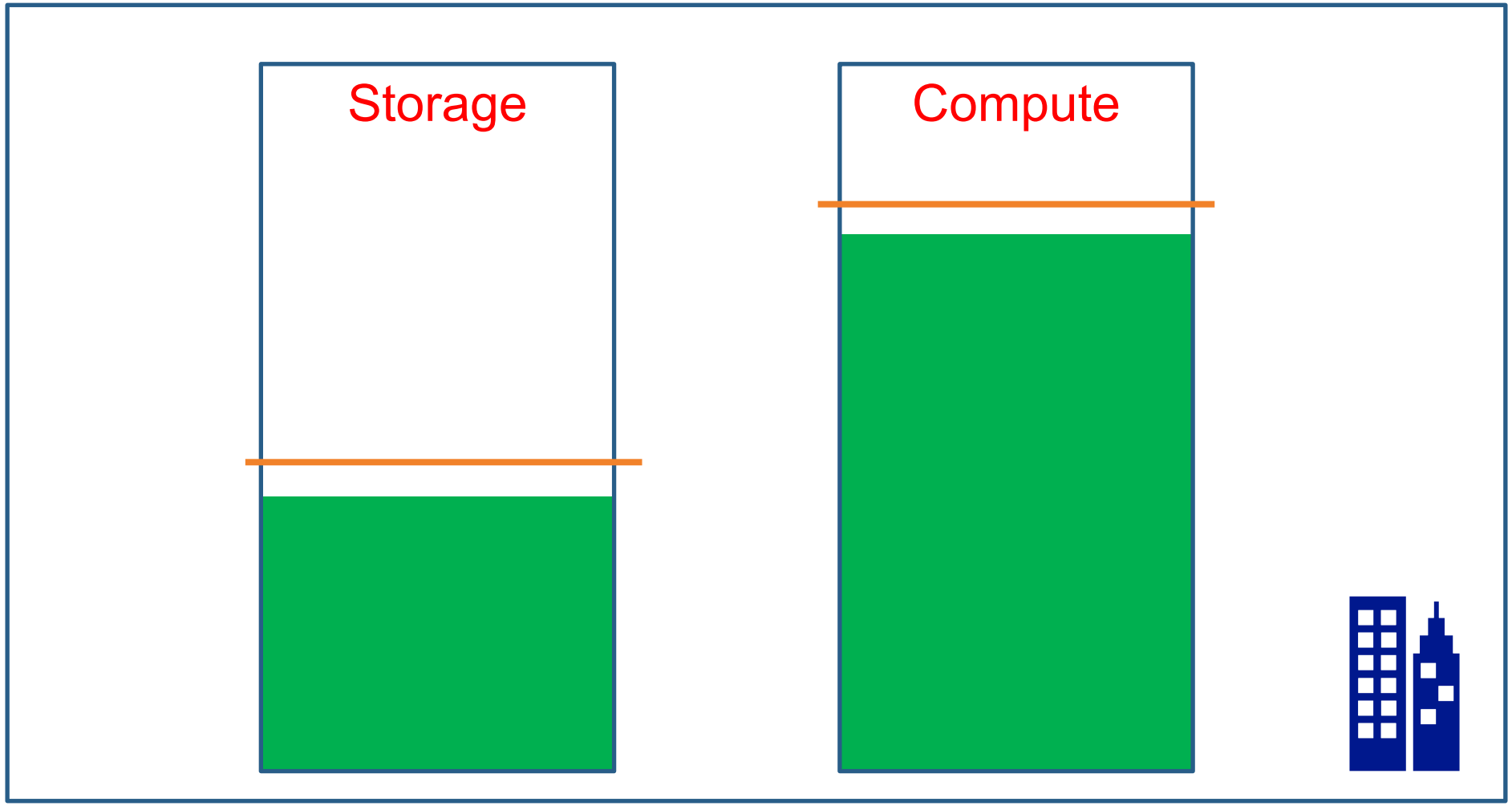

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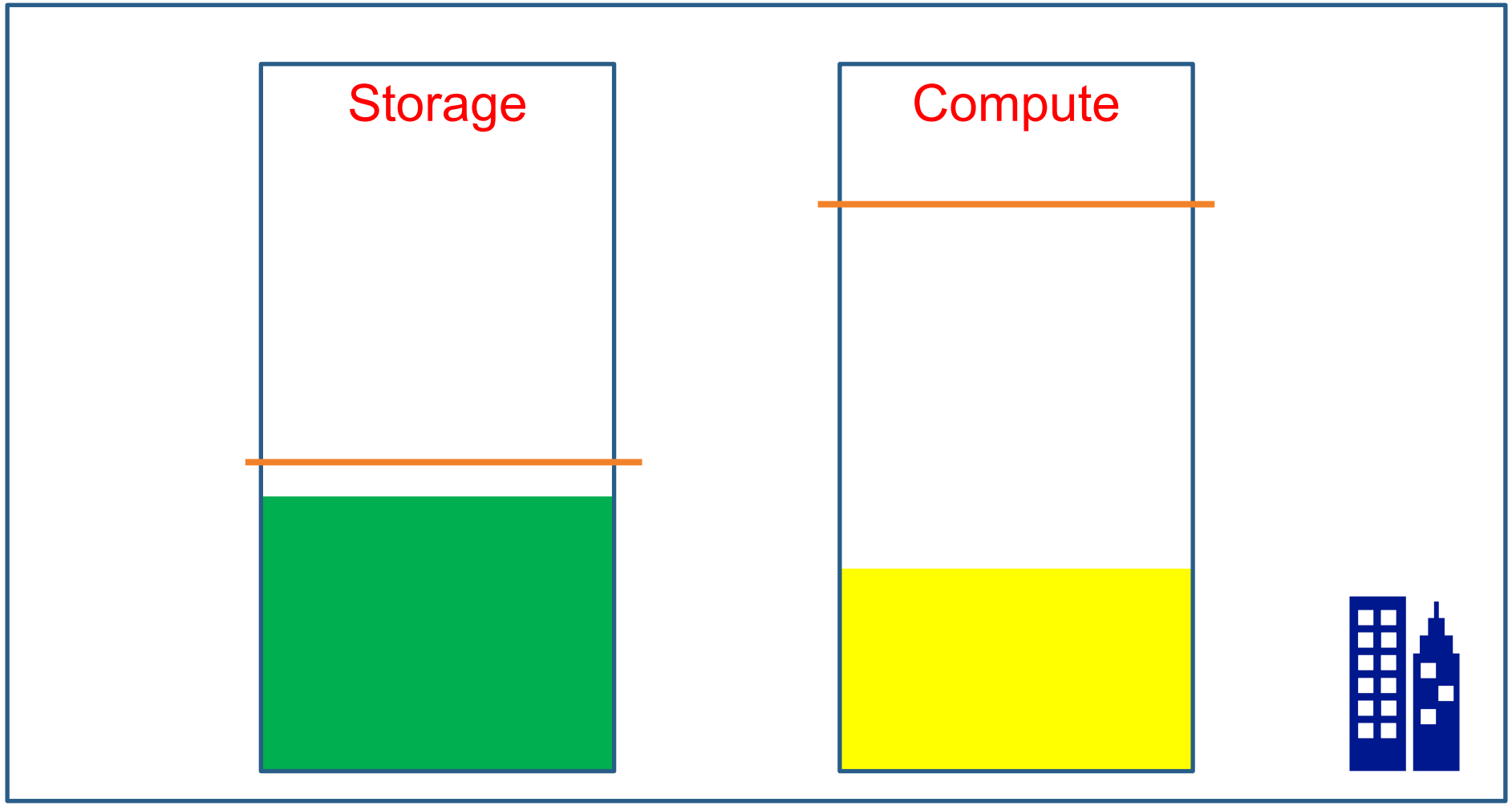

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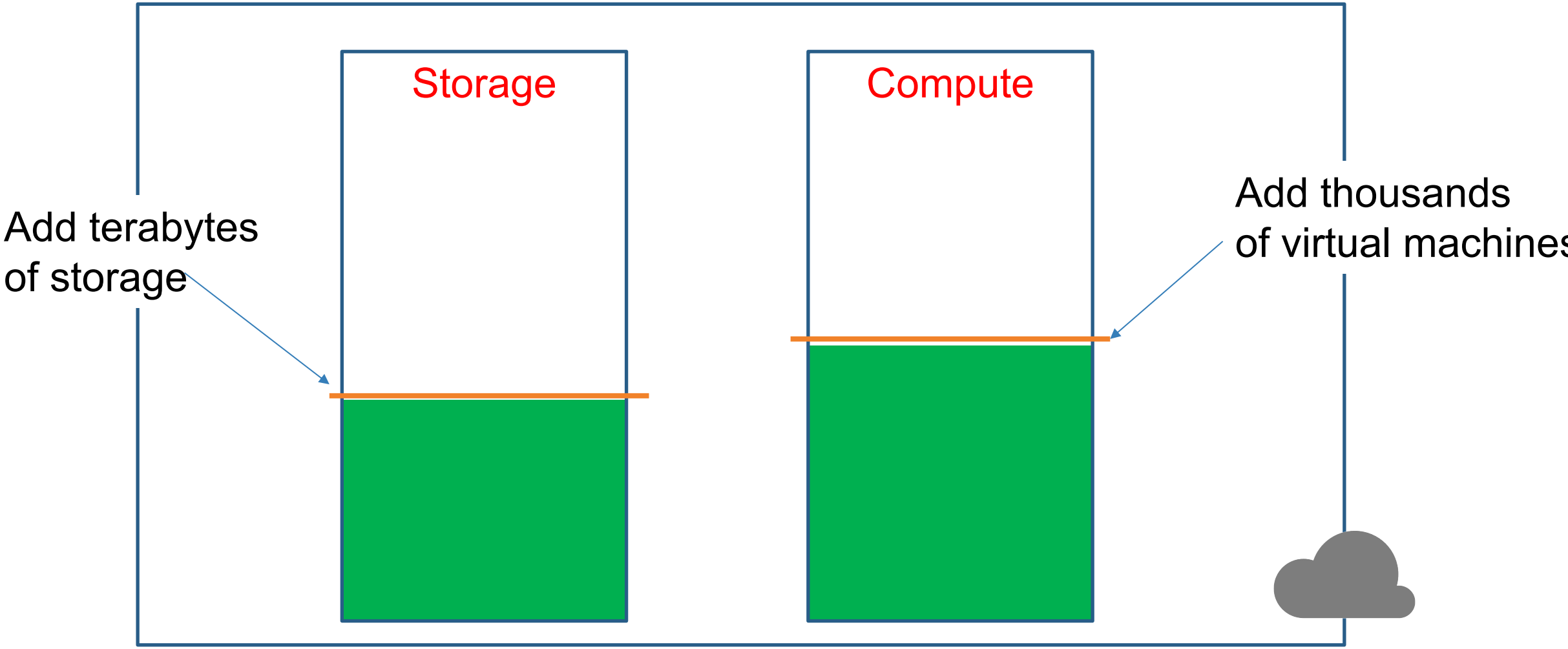


CapEx vs. OpEx

- On-premises capacity expansion = capital expense
  - Purchase equipment and licensing up-front
  - Depreciate and replace equipment
  - Renew licenses
- Cloud-base capacity expansion = operational expense
  - Billed monthly for what is used
  - No equipment purchase
  - May or may not require license purchase
- Capacity reduction
  - On-premises – possibly sell excess equipment
  - Cloud – reduce monthly cost

Consumption-Based Spending

- Capacity-based spending
  - On-premises resources
  - Some cloud resources – virtual machines
- Consumption-based spending
  - Pay only for what is used
  - Functions, Lambda
  - Services
  - Storage

Functional Advantages

- Provision environments in minutes rather than days, weeks, or months
  - No capital equipment purchases
  - Streamlined provisioning process
- Built-in access and allocation management
- Reduced administrative overhead

Also, Maybe Not

- Existing investment
- On-going operational expenses
- Data fencing
- Regulatory compliance

Managing Cloud Resources

Managing Cloud Resources

+ Cloud Management Tools
+ Web-based Cloud Management
+ Command Line Cloud Management
+ Cloud API
+ Demo: Managing Clouds

Cloud Management Tools

Web - based

{ REST }

REST API

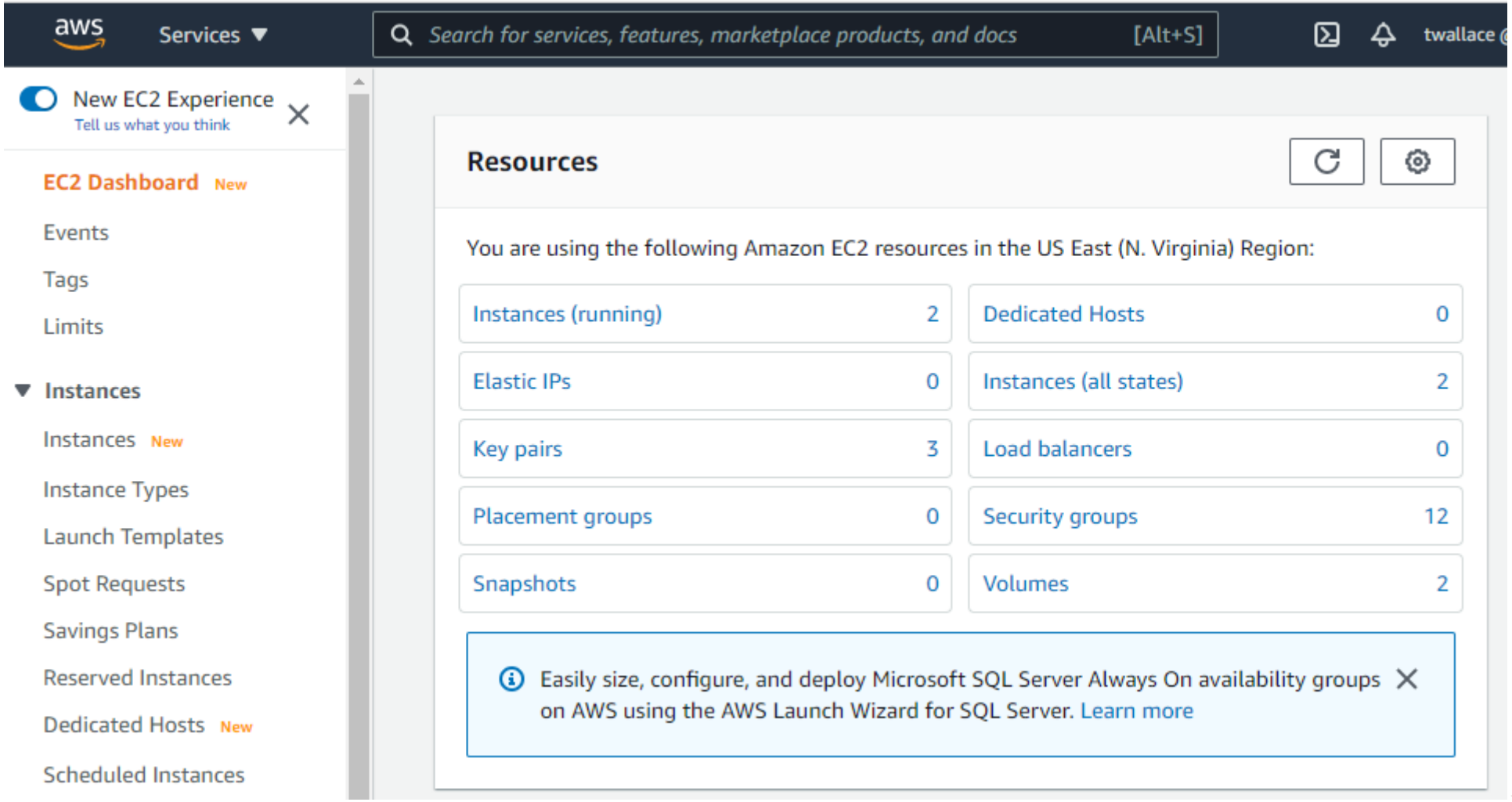
Command Line

```
C:\Program Files\Amazon\AWSCLI>aws configure
AWS Access Key ID [****************[redacted]]: [redacted]
AWS Secret Access Key [****************[redacted]]: [redacted]
Default region name [ap-southeast-2]: us-east-1
Default output format [None]:

C:\Program Files\Amazon\AWSCLI>aws s3 ls s3://popitree-bucket
2017-10-17 09:35:56         50 Stop.txt

C:\Program Files\Amazon\AWSCLI>aws s3 ls s3://popitree-sysdney-bucket

[SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed (_ssl.c:581)
```

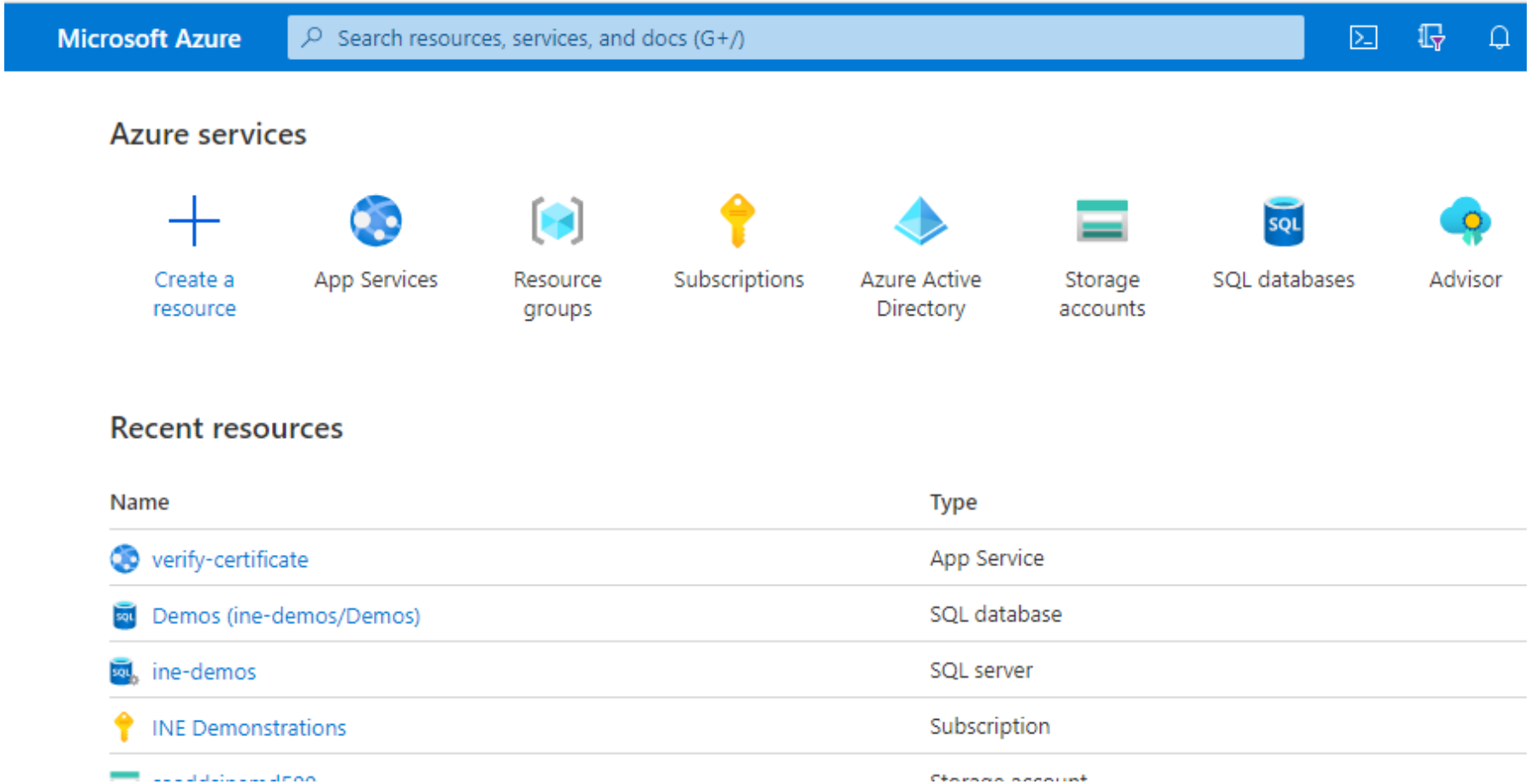
Web-based Cloud Management

- Azure: https://portal.azure.com
- AWS: https://console.aws.amazon.com/
- GCP: https://console.cloud.google.com/

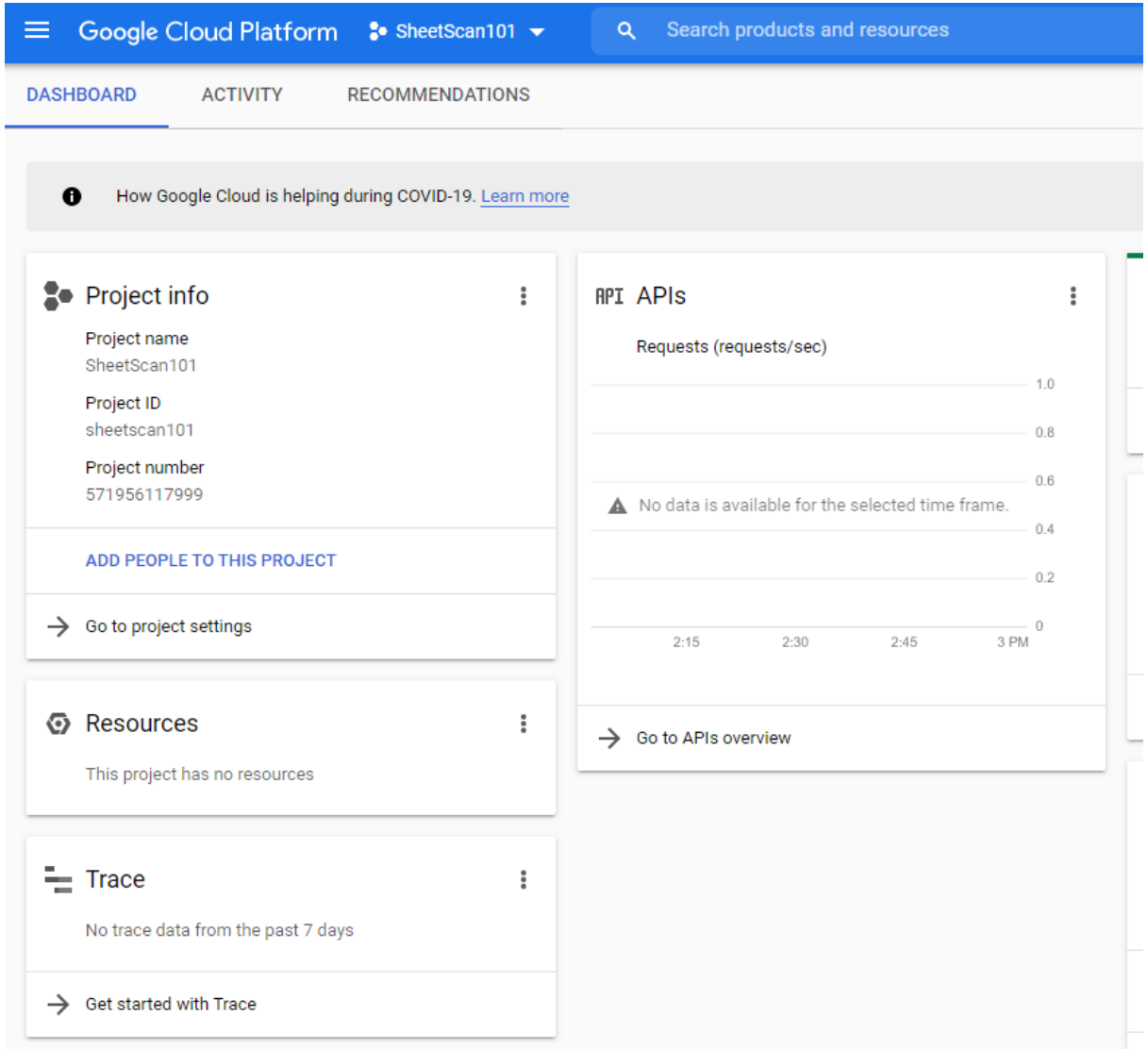
Web-based Cloud Management



Web-based Cloud Management



Web-based Cloud Management



Command Line Cloud Management

- AWS
  - AWS CLI - https://docs.aws.amazon.com/cli/latest/userguide/install-cliv2.html
  - PowerShell - https://docs.aws.amazon.com/powershell/latest/userguide/pstools-getting-set-up.html
- Azure
  - Azure CLI - https://docs.microsoft.com/en-us/cli/azure/install-azure-cli
  - Azure PowerShell `Install-Module -Name Az -AllowClobber -Scope CurrentUser`
- Google Cloud
  - Google Cloud SDK
  - Component management (kubectl)
  - Cloud Tools for PowerShell - https://cloud.google.com/tools/powershell/docs/quickstart
- Cloud shell

Cloud API

POST https://login.microsoftonline.com/{{TenantID}}/oauth2/token

Params Auth Headers (11) Body Pre-req. Tests Settings

x-www-form-urlencoded

	KEY	VALUE	DESCR
☑	grant_type	client_credentials	
☑	client_id	{{AppID}}	
☑	client_secret	{{Secret}}	
☑	resource	https://storage.azure.com/	
	Key	Value	Descr

Demo: Managing Clouds

Cloud Cost Management

Cloud Cost Management

- Cloud Pricing Models
- Demo: Calculate Resource Costs
- Cloud Billing
- Cost Monitoring
- Cost Optimization
- Demo: Optimize Storage Cost

Cloud Pricing Models

Capacity	Consumption

Beware of data transfer and transactional costs

Demo: Calculate Resource Costs

Cloud Billing

- Billing entity
- Billing cycle
- Billing management
- Billing rate
- Marketplace billing

Cost Monitoring

- Budgets
- Alerts
- Monitoring tools
- 3rd party

Cost Optimization

- Agents
  - Azure advisors
  - AWS cost anomaly detection
  - Google recommender
- Sizing
- Autoscale
- "Serverless" options
- Long-term commitments

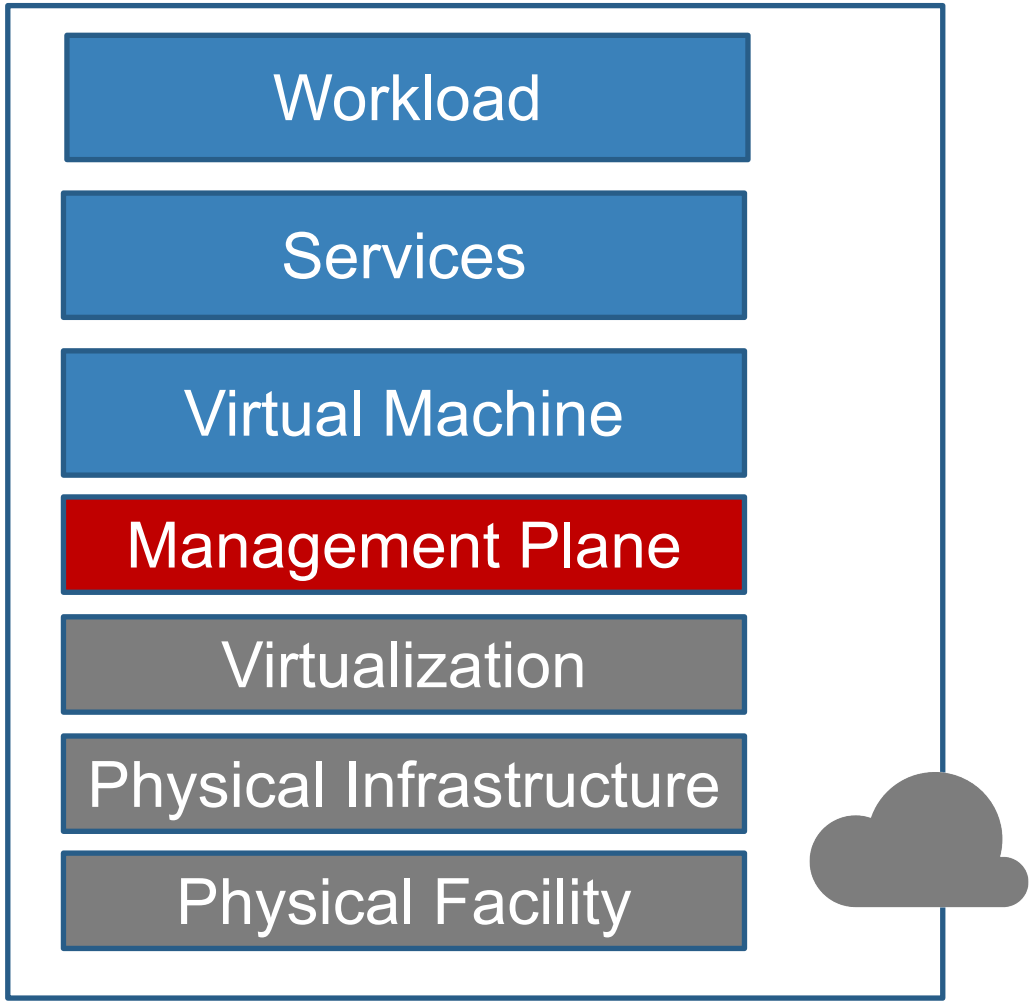
Demo: Optimize Storage Cost

Cloud Support and SLAs

Cloud Support and SLAs

- Cloud Resource Responsibility
- Cloud SLAs
- Demo: Cloud SLAs
- Cloud Support
- Demo: Cloud Support

Cloud Resource Responsibility



Cloud SLAs

- AWS
  - Baseline EC2 – 99.99% Uptime
  - AWS Service Level Agreements (SLAs) (amazon.com)
- Azure
  - Baseline virtual machine – 99.99% uptime*
  - Service Level Agreements - Home | Microsoft Azure
- Google Cloud Platform
  - Baseline virtual machine – 99.99% uptime*
  - Google Cloud Platform Service Level Agreements

**Azure and Google virtual machines must have multiple instances in different availability zones to qualify for 99.99% SLA.*

Demo: Cloud SLAs

Cloud Support

- AWS
  - Basic
  - Developer
  - Business
  - Enterprise
- Azure
  - Basic
  - Developer
  - Standard
  - Professional direct
- Google Cloud Platform
  - Basic
  - Developer
  - Production
  - Premium

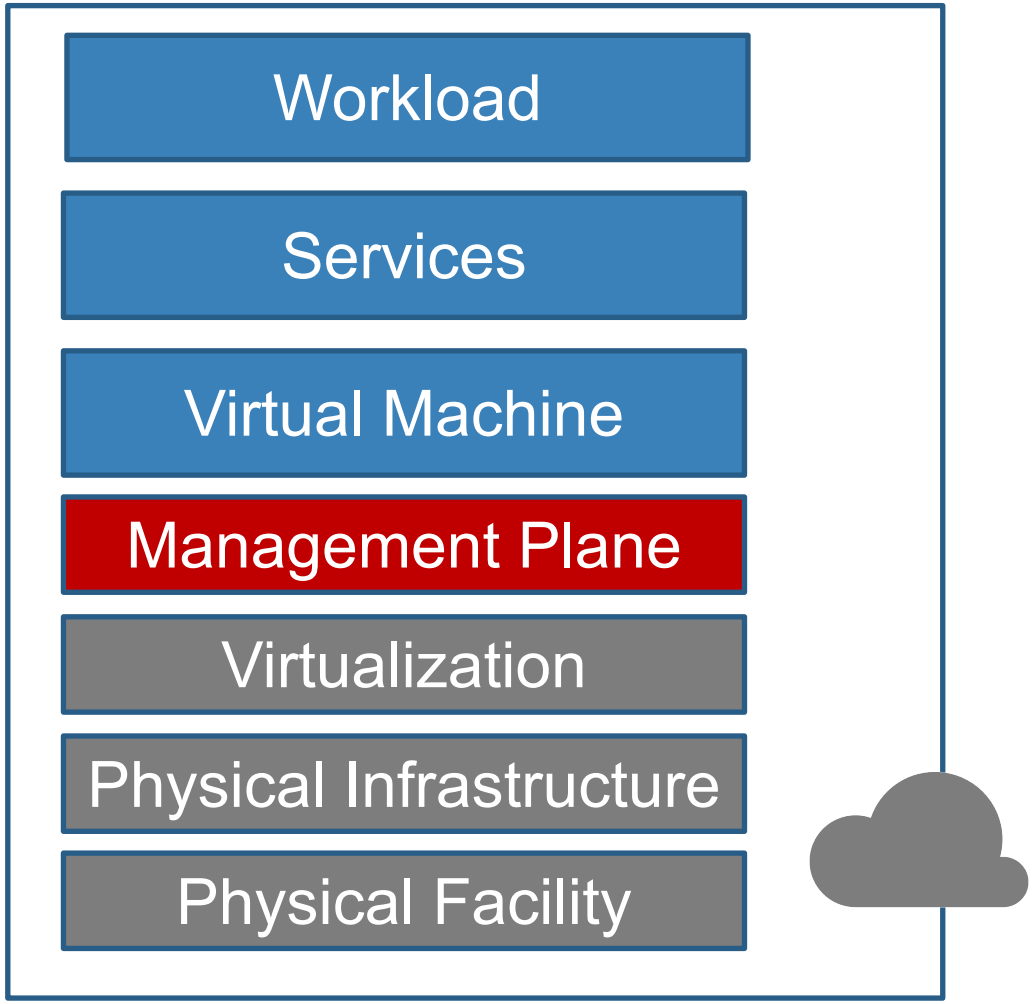
Demo: Cloud Support

Cloud Infrastructure Services

Cloud Infrastructure Services

- Infrastructure as a Service
- Cloud Networking
- Cloud Compute
- Cloud Storage
- Demo: IaaS

Infrastructure as a Service



Infrastructure as a Service

Cloud Networking

Cloud Compute

Cloud Storage

- AWS
  - S3 buckets
  - EFS
  - EBS
- Azure
  - Storage account
  - Managed disks
- Google Cloud
  - Storage buckets
  - Compute engine disks and images

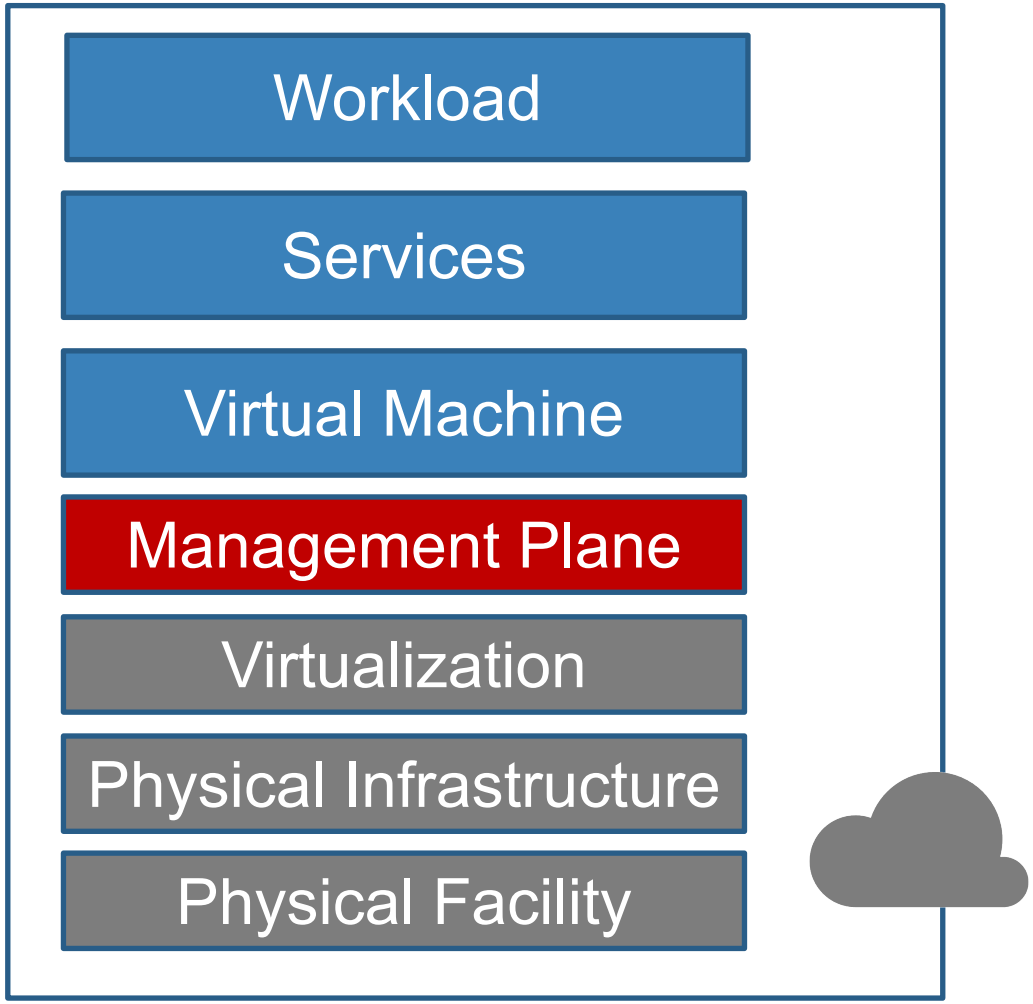
Demo: IaaS

Cloud Platform Services

Cloud Platform Services

- Platform as a Service
- Application Hosting
- Data Hosting
- Other Services
- Demo: Application Hosting

Platform as a Service



Application Hosting

AWS	Azure	Google Cloud
Lightsail Batch Lambda Beanstalk	Web Apps API Apps Function Apps Logic Apps Batch	App Engine Cloud Functions Workflows
Elastic Container Service Elastic Container Registry Elastic Kubernetes Service Fargate	Azure Container Registry Azure Container Instances Azure Container Services	Cloud Run Container Registry Kubernetes Engine Cloud Build Cloud Code

Data Hosting

AWS	Azure	Google Cloud
Amazon Aurora Amazon RDS	Azure SQL database Azure Database for PostgreSQL Azure Database for MySQL Azure Database for MariaDB	Cloud Spanner Cloud SQL
Amazon Redshift DynamoDB Amazon Elasticache Amazon DocumentDB Amazon Keyspaces Amazon Neptune	Cosmos DB Azure Synapse Azure Redis Cache	Cloud Bigtable Firestore Bigquery Firebase Memorystore

Other Services

- Authentication and identity services
- Security services
- Media services
- Migration services
- Archiving services
- Machine learning
- Cognitive services
- IoT services
- More…

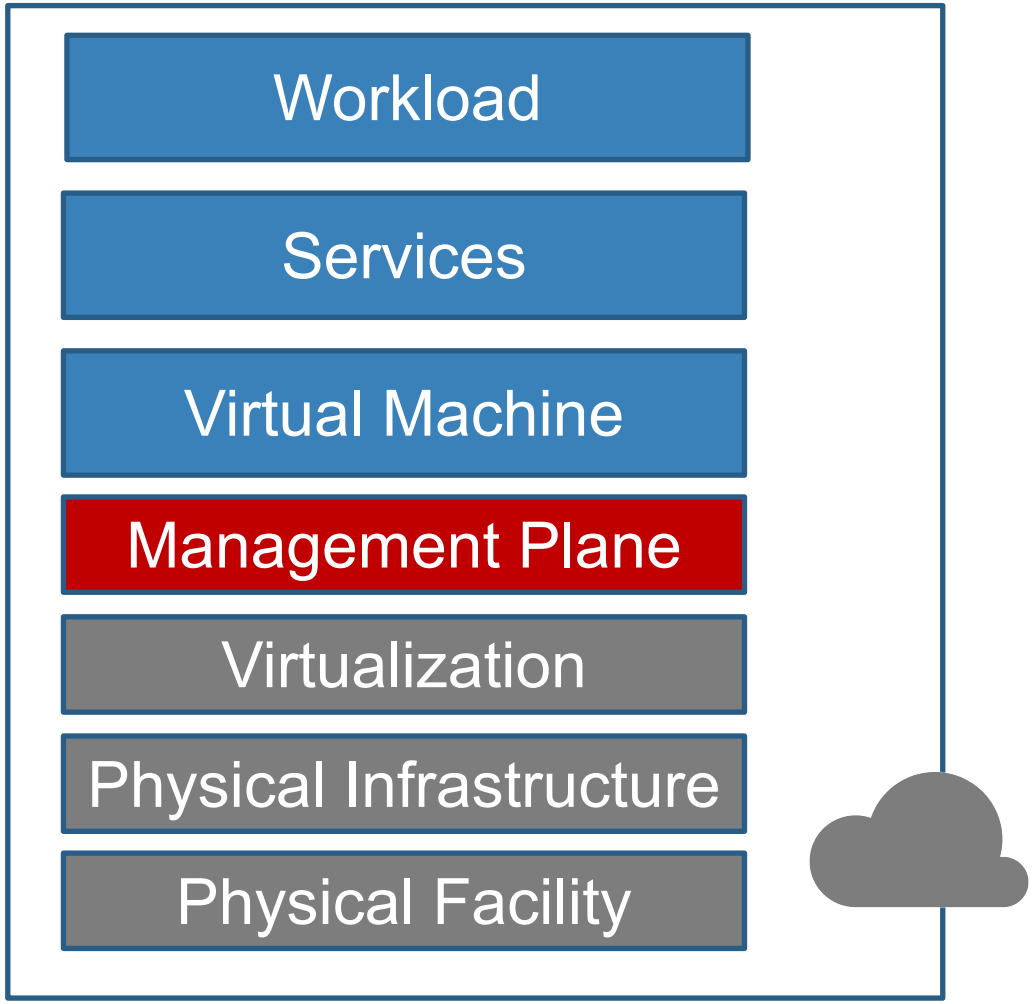
Demo: Application Hosting

Cloud Application Services

Cloud Application Services

- Software as a Service
- SalesForce
- Microsoft 365
- G Suite
- Collaboration
- Others
- Demo: Cloud Productivity

Software as a Service



SalesForce

- The leader in customer relations management (CRM)
- Depending on approach, the biggest SaaS provider
- Integration through Lightning platform
- Collaboration, coaching, and training management
- Online marketplace

Microsoft 365

- Productivity suite
  - Email
  - Collaboration
  - File management
- Online and offline versions of Office products
- CRM and business process management through Dynamics 365
- Low-code and no-code customization

G Suite

- Productivity suite
  - Email
  - Collaboration
  - File management
- Highly integrated

Collaboration

- Slack
- Zoom
- Microsoft Teams
- Google
- Others

Others

- Customer relations management
- Enterprise resource planning
- Content management systems
- Accounting
- Business continuity
- Many more

Demo: Cloud Productivity

Scalability and Availability

Scalability and Availability

- Regional Computing
- Cloud Scale
- Cloud Availability
- Demo: Cloud Scale

Regional Computing

Cloud Scale

Cloud Availability

Demo: Cloud Scale

Azure in Under 10 Minutes

Azure in Under 10 Minutes

- Demo: Azure in Under 10 Minutes

AWS in Under 10 Minutes

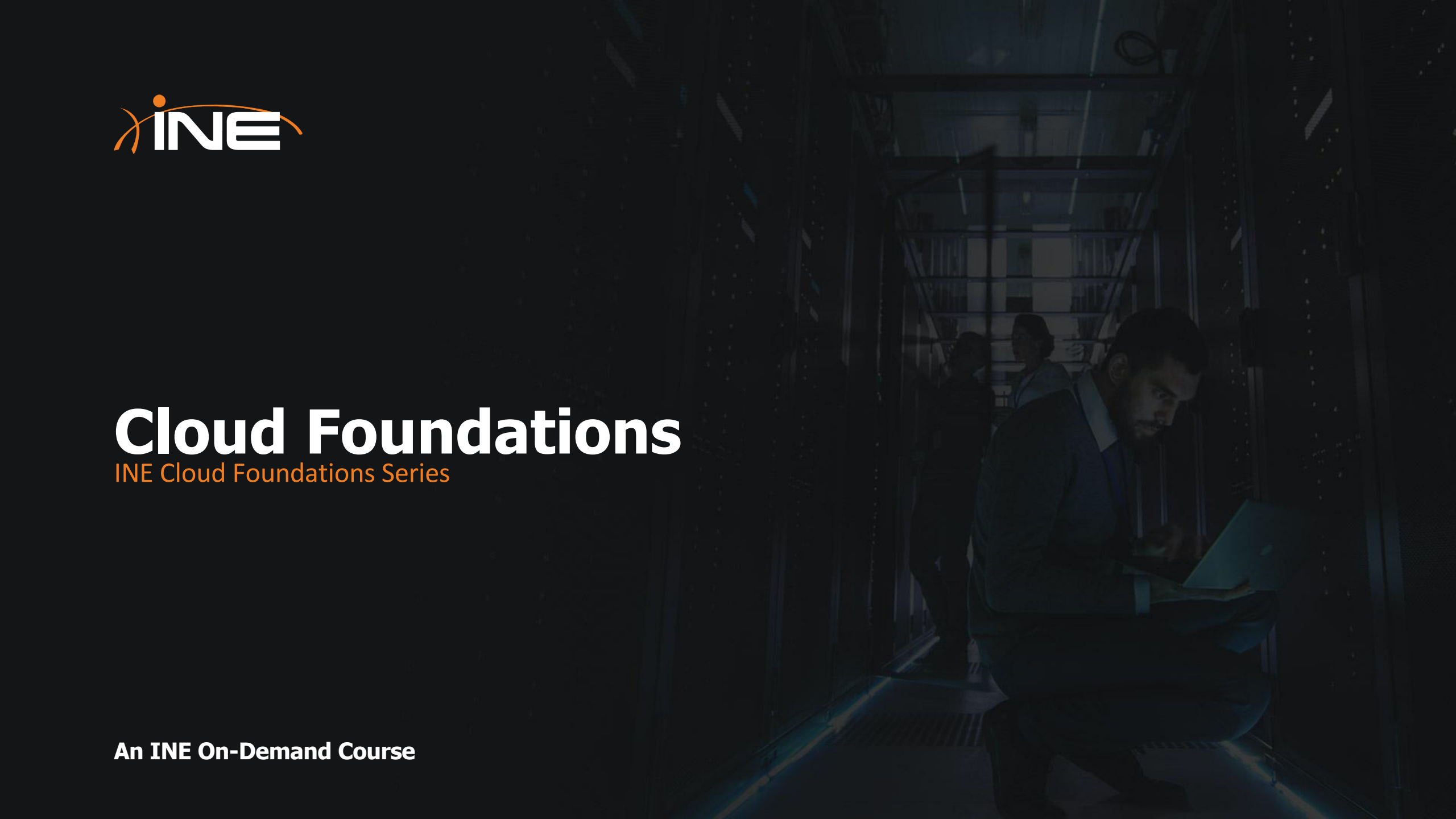
AWS in Under 10 Minutes

+ Demo: AWS in Under 10 Minutes

Google Cloud Platform in Under 10 Minutes

Google Cloud Platform in Under 10 Minutes

+ Demo: Google Cloud Platform in Under 10 Minutes



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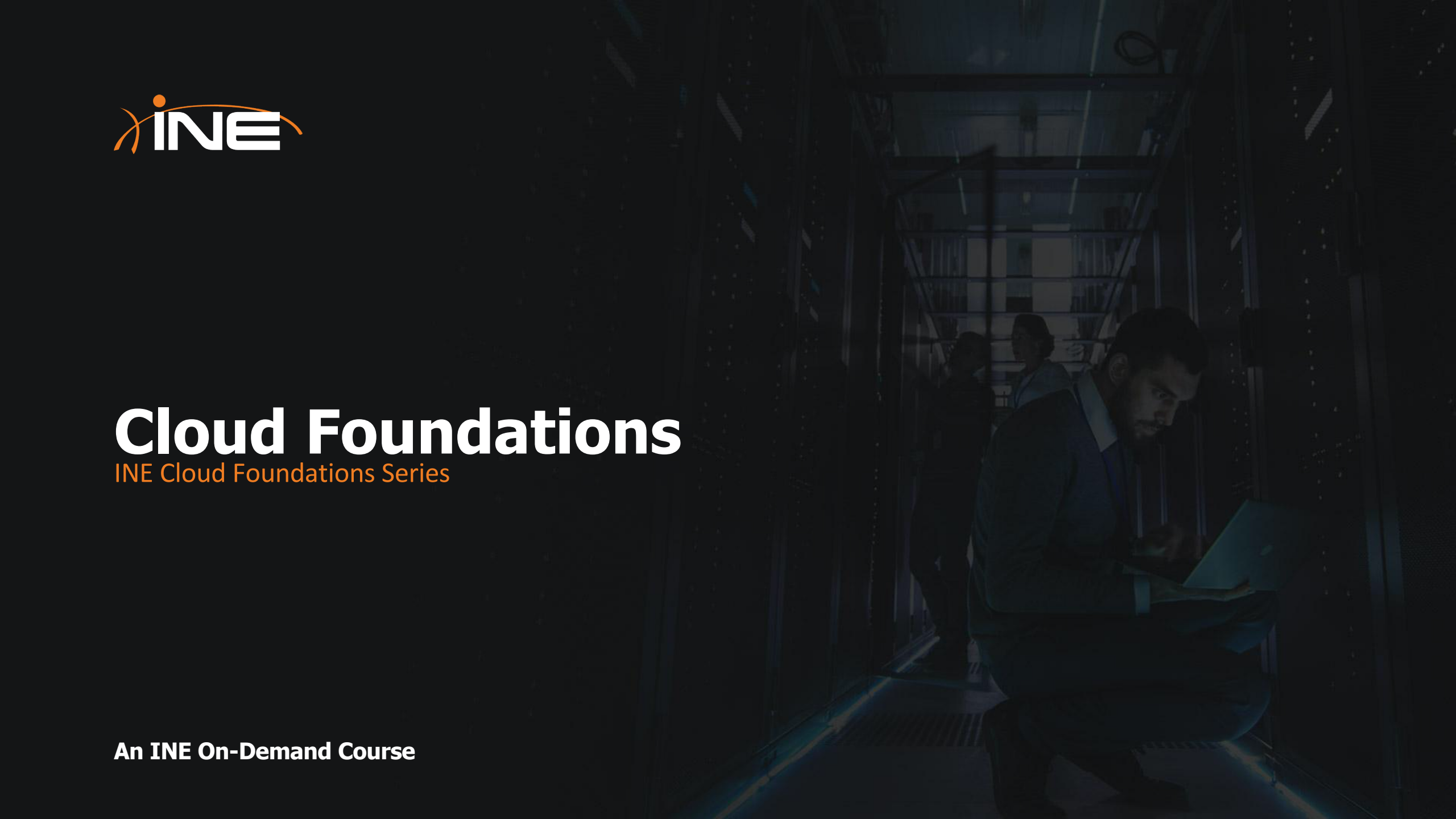
Cloud Foundations

Introduction
Cloud Management
Cloud Services
Cloud Providers



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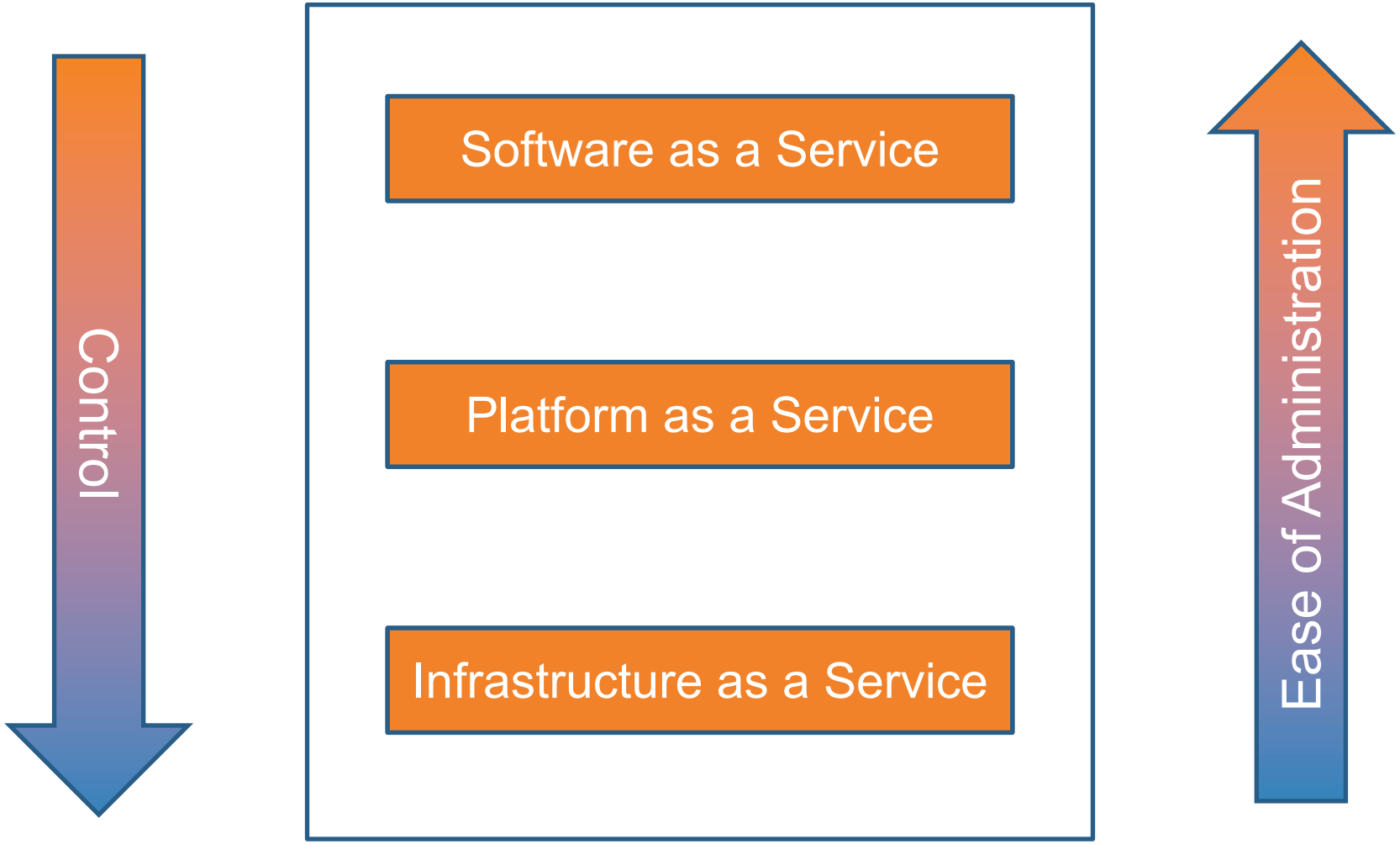
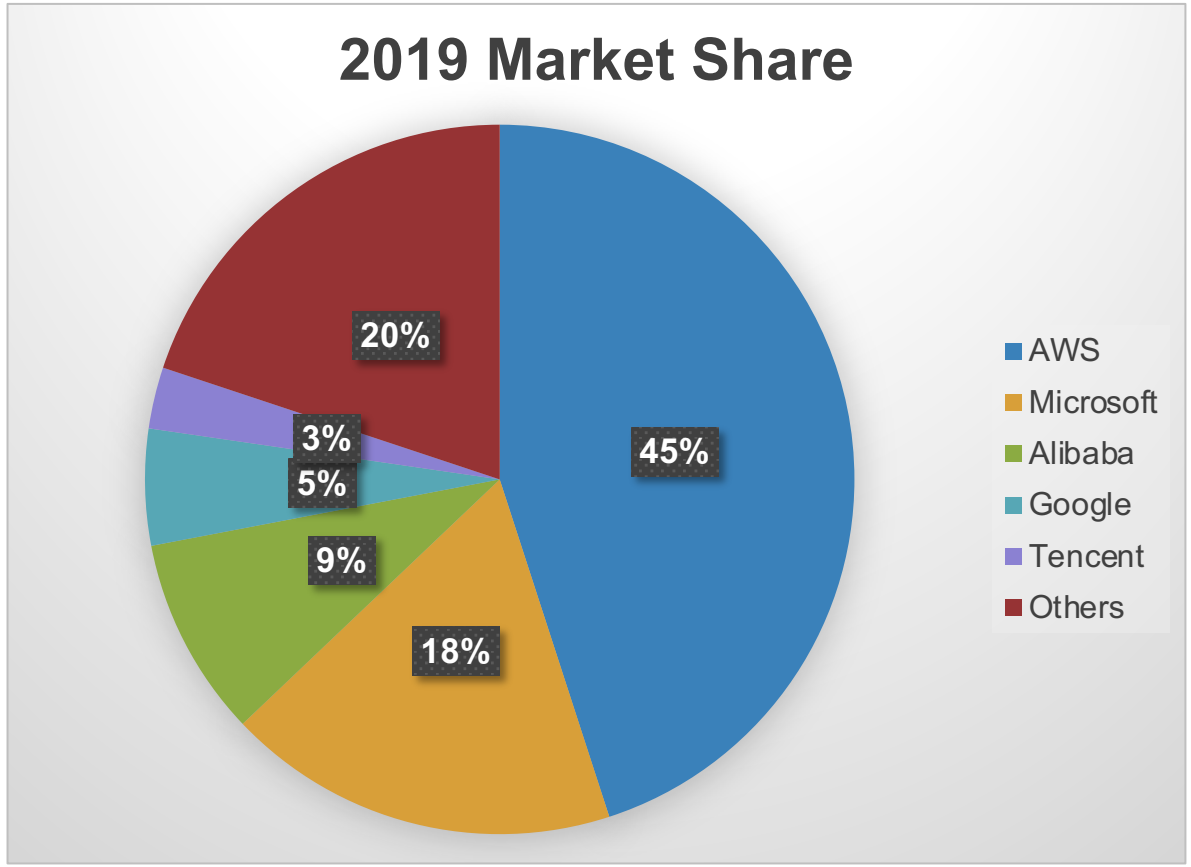
Types of Cloud Services



Cloud Market



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Cloud Management Tools

Web - based

{ REST }

REST API

Command Line

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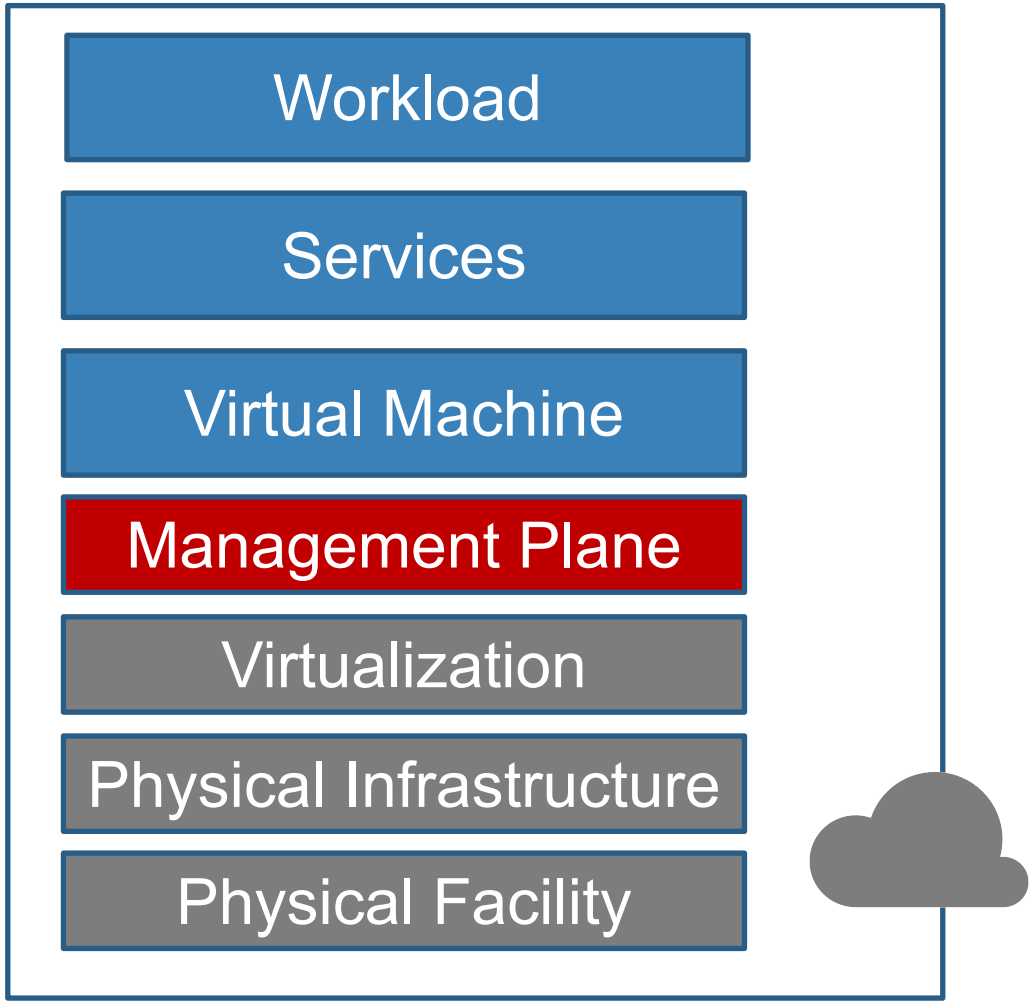
Cloud Pricing Models

Capacity	Consumption

Beware of data transfer and transactional costs

Cloud Resource Responsibility



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